

Hill Cipher:- (Substitution cipher)

- Polyalphabetic cipher (can use different cipher letter for same plaintext letter)
- it encrypts a group of letters of plaintext (like in Playfair cipher, it was encrypting a pair of letters, called digraphs such as 'ab').
- So, similarly here we will encrypt digraph, trigram etc.
- This method makes use of matrices.

Encryption:-

uses formula

$$C = KP \pmod{26}$$

where, C is Cipher Text, K is key, P is our plaintext and $\pmod{26}$ is applied because we already know there are only 26 alphabets in English language.

Step 1: Choose a key

- a key is decided firstly, but key must be a square matrix like 2×2 , 3×3 etc
- we can take any word as key, but remember that when key alphabets are placed in a matrix, it must form a square matrix like if we take "Hill" as key then placing it in matrix should make a square matrix, $\begin{bmatrix} H & i \\ l & l \end{bmatrix}$, it forms a 2×2 matrix.

Step 2: replace key alphabets with numbers.

we assign all alphabets numbers from 0-25
e.g. $A=0$, $B=1$ $Y=24$, $Z=25$.

So replace key alphabet in matrix with respective assigned number.

Step 3: Converting your plaintext in vectors

- Since, this cipher works with matrices so we also need to convert our plaintext into vectors of $n \times 1$.
- If we have a plaintext like "Attack" which we have to encrypt then it will be represented in vectors based on key square now.
- If key size is 2×2 matrix then plaintext vectors would be in 2×1 vectors.
- You should consider it generally as, if we have key matrix size as $n \times n$, then plaintext vectors should be $n \times 1$.

Step 4: multiplication of key matrix and plaintext vector

- we'll start putting values in formula

$$C = KP \text{ mod } 26$$

where every plaintext vector will be multiplied to key matrix and then mod 26 will be calculated, to provide cipher alphabets for those respective plaintext alphabets.

- we will continue multiplying every plaintext vector with key to obtain our ciphertext until all plaintext are multiplied and we have got final ciphertext.

Decryption:

uses formula

$$P = K^{-1}C \text{ mod } 26$$

it uses inverse of key matrix to decrypt

Step 1: Firstly we need to calculate K^{-1} , and then rest of the process will be same like encryption.

→ To calculate K^{-1} , we need to find

$$K^{-1} = \frac{1}{|K|} \cdot \text{adj}(K)$$

(1) we will find $|K|$ like if we have to calculate $|K|$ of matrix $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ then $|K|$ would be $(ad - bc)$

(2) Then find multiplicative inverse of $|K|$

$$\text{i.e. } |K| \cdot K^{-1} = 1 \text{ mod } 26$$

$$|K| \cdot K^{-1} \text{ mod } 26 = 1$$

(we need to multiply our $|K|$ value with a number that provides answer as 1, when mod 26 is applied like if we have got determinant $|K|$ value as 3 then we should do it as;

(can take K^{-1} value as 9 so it will provide ans 1)

$$\begin{aligned} |K| \cdot K^{-1} \text{ mod } 26 &= 1 \\ 3 \cdot K^{-1} \text{ mod } 26 &= 1 \\ 3 \cdot 9 \text{ mod } 26 &= 1 \\ 27 \text{ mod } 26 &= 1 \\ 1 &= 1 \end{aligned}$$

(3) Now, we will find adj of k , so
let a matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then

$$\text{adj}(A) \text{ will be } \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

But remember, we have to remove negative values before decryption, so to remove negative values we will add 26 to them.

Step 2: Now when we have $|k|$ and $\text{adj}(k)$
∵ Since $\frac{1}{|k|}$ is equal to k^{-1}

So we will use k^{-1} value next helped us achieve answer as 1.

Multiply k^{-1} with $\text{adj}(K)$ and apply mod 26 to its answer.

Step 4: Now reapply encryption process, by converting our ciphertext into vectors and multiplying it with k^{-1} to get plaintext message.

Example:

Plaintext = "Attack"

so 2x2 matrix of key

key in numeric form
key = $\begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$

A 0	E 4	I 8	M 12	Q 16	U 20	Y 24
B 1	F 5	J 9	N 13	R 17	V 21	Z 25
C 2	G 6	K 10	O 14	S 18	W 22	
D 3	H 7	L 11	P 15	T 19	X 23	

As we have key size 2x2, so

Converting plaintext into 2x1 vectors

Then "Attack" will be

$$\begin{bmatrix} A \\ T \end{bmatrix} \begin{bmatrix} T \\ A \end{bmatrix} \begin{bmatrix} C \\ K \end{bmatrix}$$

Encryption:

1. Take first vector and convert it into numeric form and multiply with key

$$C = kp \text{ mod } 26$$

$$1^{\text{st}} \text{ vector} = \begin{bmatrix} A \\ T \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 19 \end{bmatrix}$$

$$C = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \times \begin{bmatrix} 0 \\ 19 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 2 \times 0 + 3 \times 19 \\ 3 \times 0 + 6 \times 19 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 57 \\ 114 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

Corresponding alphabets = $\begin{bmatrix} F \\ K \end{bmatrix}$

First vector
Cipher Text
Fk

2nd vector $\begin{bmatrix} T \\ A \end{bmatrix} = \begin{bmatrix} 19 \\ 0 \end{bmatrix}$

$$\begin{aligned} C &= kp \pmod{26} \\ &= \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 19 \\ 0 \end{bmatrix} \pmod{26} \\ &= \begin{bmatrix} 2(19) + 3(0) \\ 3(19) + 6(0) \end{bmatrix} \pmod{26} \\ &= \begin{bmatrix} 38 \\ 57 \end{bmatrix} \pmod{26} \\ &= \begin{bmatrix} 12 \\ 5 \end{bmatrix} \quad \text{Corresponding letter} \quad \begin{bmatrix} M \\ F \end{bmatrix} \end{aligned}$$

So TA becomes MF

lastly 3rd vector $\begin{bmatrix} C \\ k \end{bmatrix} \rightarrow \begin{bmatrix} 2 \\ 10 \end{bmatrix}$

$$\begin{aligned} C &= kp \pmod{26} \\ &= \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 10 \end{bmatrix} \pmod{26} \\ &= \begin{bmatrix} 2(2) + 3(10) \\ 3(2) + 6(10) \end{bmatrix} \pmod{26} \\ &= \begin{bmatrix} 34 \\ 66 \end{bmatrix} \pmod{26} \\ &= \begin{bmatrix} 8 \\ 14 \end{bmatrix} \rightarrow \begin{bmatrix} I \\ O \end{bmatrix} \end{aligned}$$

Ck becomes IO

So

Plain Text = ATTACK
Cipher Text = FKMFI O

Decryption

Cipher Text = FKMFI O

Firstly make vectors of cipher Text as

$$\begin{bmatrix} F \\ K \end{bmatrix} \begin{bmatrix} M \\ F \end{bmatrix} \begin{bmatrix} I \\ O \end{bmatrix}$$

now calculating $k^{-1} = \left[\frac{1}{|k|} \text{adj}(k) \right]$ ~~calculate~~

$$|k| = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$$

$$= (2)(6) - (3)(3)$$

$$= 12 - 9 = 3$$

(k^{-1}) , finding multiplicative inverse k^{-1}

$$|k| \cdot k^{-1} \mod 26 = 1$$

$$3 \cdot 9 \mod 26 = 1$$

$$27 \mod 26 = 1$$

$$1 = 1$$

So k^{-1} value is 9

$$\text{adj}(k) = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 6 & -3 \\ -3 & 2 \end{bmatrix}$$

removing negative values

$$= \begin{bmatrix} 6 & -3+26 \\ -3+26 & 2 \end{bmatrix}$$

$$\text{adj}(k) = \begin{bmatrix} 6 & 23 \\ 23 & 2 \end{bmatrix}$$

$$K^{-1} = \frac{1}{|K|} \cdot \text{adj } K$$

$$= \frac{1}{9} \begin{bmatrix} 6 & 23 \\ 23 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 54 & 207 \\ 207 & 18 \end{bmatrix}$$

find mod 26 $\rightarrow$ simplify

$$= \begin{bmatrix} 54 & 207 \\ 207 & 18 \end{bmatrix} \text{ mod } 26$$

$$K^{-1} = \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix}$$

Now, start decrypting

$$\text{First vector} = \begin{bmatrix} F \\ K \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$$P = K^{-1} C \text{ mod } 26$$

$$= \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix} \begin{bmatrix} 5 \\ 10 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 260 \\ 305 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 0 \\ 19 \end{bmatrix} \rightarrow \begin{bmatrix} A \\ T \end{bmatrix}$$

Second vector $\begin{bmatrix} M \\ F \end{bmatrix} = \begin{bmatrix} 12 \\ 5 \end{bmatrix}$

$$\begin{aligned} P &= k^{-1} C \pmod{26} \\ &= \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix} \begin{bmatrix} 12 \\ 5 \end{bmatrix} \\ &= \begin{bmatrix} 149 \\ 390 \end{bmatrix} \pmod{26} \\ &= \begin{bmatrix} 19 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} T \\ A \end{bmatrix} \end{aligned}$$

lastly, third vector $\begin{bmatrix} I \\ O \end{bmatrix} = \begin{bmatrix} 8 \\ 14 \end{bmatrix}$

$$\begin{aligned} P &= k^{-1} C \pmod{26} \\ &= \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix} \begin{bmatrix} 8 \\ 14 \end{bmatrix} \pmod{26} \\ &= \begin{bmatrix} 366 \\ 452 \end{bmatrix} \pmod{26} \\ &= \begin{bmatrix} 2 \\ 10 \end{bmatrix} \rightarrow \begin{bmatrix} C \\ K \end{bmatrix} \end{aligned}$$

Cipher Text = FKMP IO

Plaintext = Attack

Playfair Cipher

→ In Playfair Cipher, we encrypt a pair of alphabets (digraphs) instead of single alphabet.

Encryption:

Step 1: Generate the key Square (5×5)

- The key Square is a 5×5 matrix that acts as the key for encrypting the plaintext. Each of the 25 alphabets must be unique and one letter of the alphabet mostly j is omitted from table - If plaintext contains letter j then it is replaced by i .
- The initial alphabets in the key square are the unique alphabets of key in the order in which they appear followed by the remaining letters of the alphabet in order.

Step 2: making digraphs of plaintext

- The plaintext is split into pairs of two letters (digraphs) - For word 'instrument' digraphs will be 'in', 'st', 'ru', 'me', 'nt', -
- But if there are odd numbers of plaintext then an anonymous alphabet let say x is added to last letter to make pair.

like for word 'instruments' it will be as

'in', 'st', 'ou', 'me', 'nt', 'sx' (x is added for pair)

- Pair Cannot be made with same letter, we have to break digraph and add an anonymous letter. e.g

Plaintext = 'hello'

digraphs = 'he', 'lx', 'lo'

Rules for encryption:

1. If both letters are in the same row - Take the letter to the right of each one (going back to the leftmost if at the rightmost position)
2. If both the letters are in the same column. Take the letter below each one (going back to the top if at the bottom)
3. If neither of the above rule is true, then take first letter row and second word column and use the ~~word~~ letter at the intersection point or else we can say form a rectangle with two letters and take letters on the horizontal opposite corner of rectangle.

Example:

Plaintext = "Instruments"

key = "Monarchy"

Step 1: Generate key square of 5x5

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Step 2: digraphs

'In', 'st', 'ru', 'me', 'nt', 'sx'

i → G

n → A

s → T

t → L

r → M

u → Z

m → C

e → I

n → R

t → Q

s → X

x → A

Plaintext = Instruments

Ciphertext = GATLMZCLRXA

Decryption:

1 → Generate same 5x5 matrix with key.

2 → make digraphs

3 → Rules are reversed

(i) If both the letters are in the same column,

Take the letter above each one (going back to the bottom if at the top position)

(ii) If both the letters are in the same row

take the letter to the left of each one (going back to the rightmost if at the leftmost position)

(iii) If neither of above rule is true : form a rectangle with two letters and take letters on opposite horizontal corner of rectangle).

Example

Ciphertext = "GATLMZCLRQXA"

key = "monarchy"

m	a	n	a	r
c	h	y	b	d
e	f	g	i/j	k
l	p	q	s	t
u	v	w	x	z

ga → in

tl → st

mz → ru

cl → me

qr → nt

XA → BX

Polyalphabetic Cipher

Vignere Cipher

→ Encryption is done using 26×26 matrix or 26×26 Table.

Method 1:

Keys



		Plaintext → x y z			
A	B	C	D	E	F
A	A	B	C	D	E
B	B	C	D	E	F
C	C	D	E	F	G
D	D	E	F	G	H
E	E	F	G	H	I
F	F	G	H	I	J
G	G	H	I	J	K
H	H	I	J	K	L
I	I	J	K	L	M
J	J	K	L	M	N
K	K	L	M	N	O
L	L	M	N	O	P
M	M	N	O	P	Q
N	N	O	P	Q	R
O	O	P	Q	R	S
P	P	Q	R	S	T
Q	Q	R	S	T	U
R	R	S	T	U	V
S	S	T	U	V	W
T	T	U	V	W	X
U	U	V	W	X	Y
V	V	W	X	Y	Z
W	W	X	Y	Z	A
X	X	Y	Z	A	B
Y	Y	Z	A	B	C
Z	Z	A	B	C	D

We have to list key letter with Plaintext letter, and key is repeated until both the Plaintext length and key length is same.

For example : Plaintext = "Give money"

Key = "lock"

then we will do it as;

G	I	V	E	M	O	N	E	Y
L	O	C	K	L	O	C	K	L

∴ Repeating the key "lock" so that no. of letters of both plaintext and key are same.

Ciphertext = RWXOXCPQJ

⇒ We simply check through the table to find intersecting letter from table - Like firstly we will take 'L' Row and 'G' Column and intersection letter will be used as cipher letter.

Decryption:

Rows in table →	Cipher	R	W	X	O	X	C	P	O	J
	key	L	O	C	K	L	O	C	K	L
	Plaintext	G	I	V	E	M	O	N	E	Y

For decryption, we will use key letter row and find where is ciphertext letter in that row, - once found we will use that column letter as plaintext -

like for first letter, we have to check key letter 'L' Row and find letter 'R' in that row once find, we will use its first column letter as Plaintext letter which is 'G'.

vignere Cipher

Method 2.. (If ^{26x26} table is not provided)
uses formula

Encryption

$$E_i = (P_i + k_i) \bmod 26$$

Decryption

$$D_i = (C_i - k_i) \bmod 26$$

Example:

Plaintext = "She is listening"

key = "Pascal"

∴ key stream → (15, 0, 18, 2, 0, 11) - The key stream is repetition of this initial key stream (as many times as needed)

Plaintext	S	h	e	i	s	l	i	s	t	e	n	i	n	g
P's value	18	7	4	8	18	11	8	18	19	4	13	8	13	6
key stream	15	0	18	2	0	11	15	0	18	2	0	11	15	0
key stream														
C's value	7	7	22	10	18	22	23	18	11	6	13	19	2	6
Ciphertext	H	H	W	K	S	W	X	G	G	G	T	C	G	

Use same table for decryption process
∴ during decryption if $(C_i - k_i)$ results in negative value, then add 26 to it, so it will be positive.